

Lectures 14

In Class Exercises

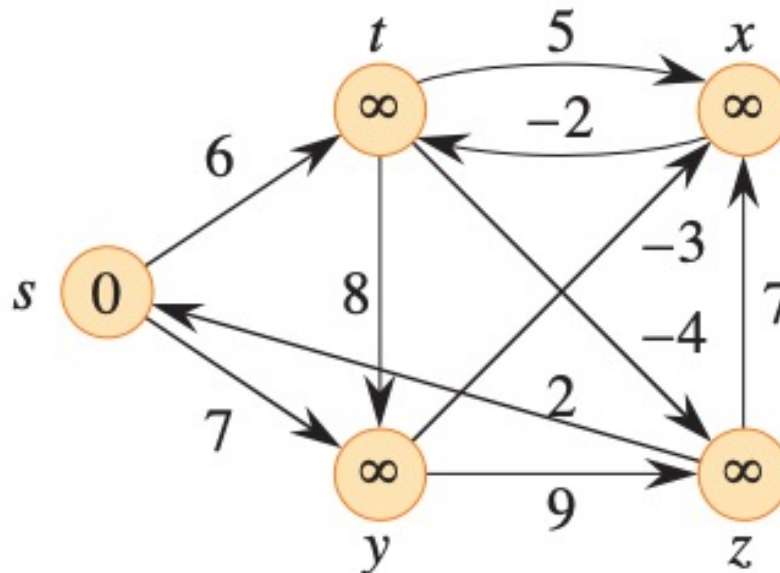


Exercise 1

Run the Bellman-Ford algorithm on the directed graph below, using vertex **z** as the source. In each pass, relax the edges in the following order:

$(t,x), (t,y), (t,z), (x,t), (y,x), (y,z), (z,x), (z,s), (s,t), (s,y)$.

After each pass show the d and π values.

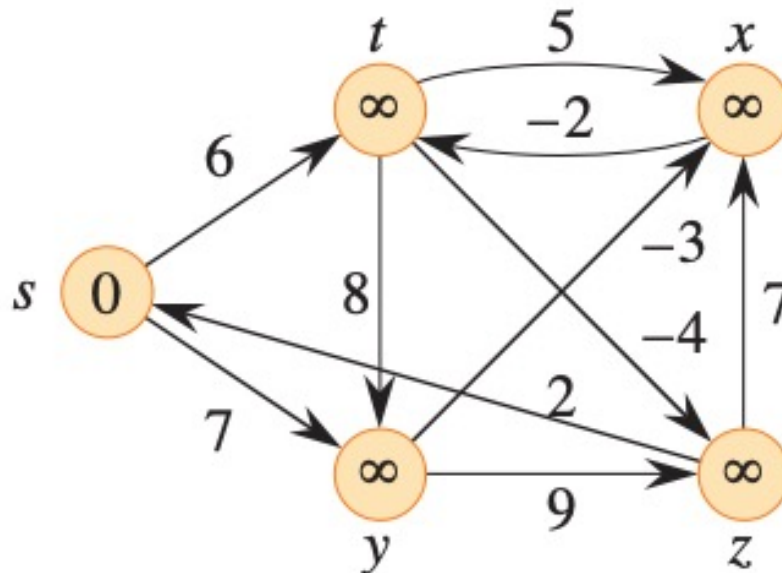


Exercise 2

Run the Bellman-Ford algorithm on the directed graph below, using vertex **x** as the source. In each pass, relax the edges in the following order:

$(t,x), (t,y), (t,z), (x,t), (y,x), (y,z), (z,x), (z,s), (s,t), (s,y)$.

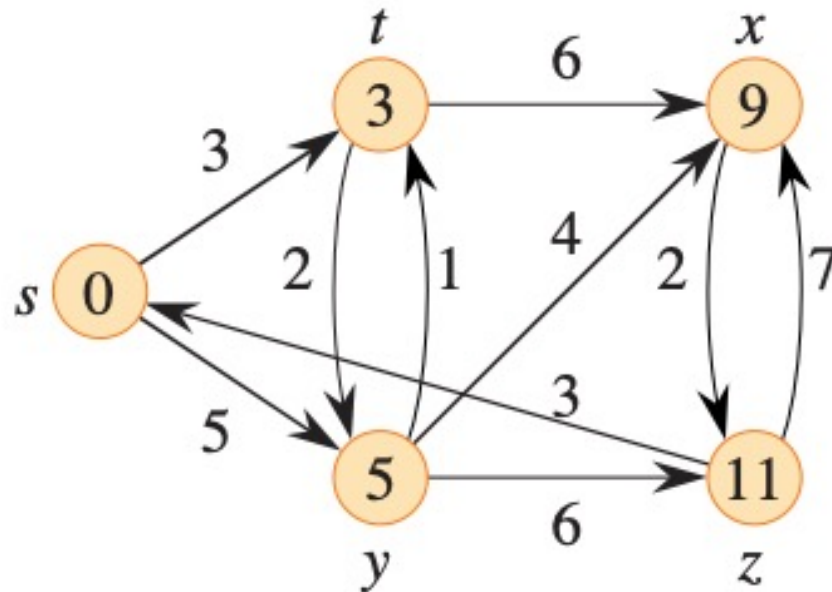
After each pass show the d and π values.



Exercise 3

Run the Dijkstra's algorithm on the directed graph below, using vertex s as the source.

After each iteration of the while loop, show the d and π values and the vertices in the set S .



Exercise 4

Run the Dijkstra's algorithm on the directed graph below, using vertex **z** as the source.

After each iteration of the while loop, show the d and π values and the vertices in the set S .

